

Seungheon Lee

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Work Experience

Research Engineer

HYUNDAI MOTOR COMPANY

Hwaseong, S.Korea

Jan. 2025 - Present

- Managing the Robotaxi project, overseeing timelines and team coordination.

Software Engineer Intern

SAMSUNG RESEARCH

Seoul, S.Korea

Jul. 2024 - Aug. 2024

- Developed a monitoring system for LLM API servers, addressing previously unimplemented functionalities for an upcoming service launch.
- Built Kubernetes clusters and set up a Dev server to support scalable infrastructure and system development.
- Collaborated with administrators to identify and implement key features, including detailed API analytics and real-time alert systems.

Undergraduate Researcher

ROBOTICS LAB, UNIVERSITY OF SEOUL

Seoul, S.Korea

Mar. 2022 - May. 2024

- Presented 3 posters and oral papers at academic conferences.
- Participated in 8 robotics competitions, gaining hands-on experience in problem-solving and teamwork.

Education

University of Seoul

B.S. IN MECHANICAL INFORMATION ENGINEERING, TOP GRADUATE

Seoul, S.Korea

Mar. 2019 - Feb. 2025

- GPA: 4.15 / 4.5 (Major GPA: 4.35 / 4.5)

Skills

Programming Languages

Python, C/C++

Software/Tools

Linux, Kubernetes, MATLAB, PyTorch, Grafana, Weights & Biases (W&B)

Hardware/Sensors

LiDAR (2D, 3D), GPS, RGB-D Camera

Projects

Computer Vision AI Model Optimization for NPU-based Inference

Sep. 2024 - Dec. 2024

- 2-member project for the 1st AI Semiconductor Competition, collaborating with Furiosa AI.
- Achieved up to 533% improvement in NPU utilization compared to the baseline code, while increasing quantization accuracy.
- Performed MLOps-related tasks, integrating tools like WandB and MLflow into the codebase for experiment tracking and model management.

AI-powered Smart Bartending System

Mar. 2024 - Oct. 2024

- 4-member project delivering a fully integrated solution combining hardware and software.
- Built a system to recommend and prepare cocktails based on user preferences and mood via a chatbot.
- Designed and implemented hardware components for automated drink preparation.

Awards & Scholarships

AWARDS

2024 **Grand Prize (Minister of Science and ICT Award)**, 1st AI Semiconductor Competition

Seoul, S.Korea

2023 **First Place**, RoboCup Autonomous Robot Manipulation Challenge, hosted by MathWorks

Bordeaux, France

2023 **Grand Prize**, Autonomous Robot Race Competition

Cheongju, S.Korea

2023 **Excellence Award**, Engineering Research Internship Outcome Presentation

Seoul, S.Korea

SCHOLARSHIPS

2023 - 2025 **OK Scholarship**, OK Scholarship Foundation

~\$19,000

2023 - 2024 **Blue Lighthouse KB Dream Wave 2030 Promising Field Scholarship**, Korea Scholarship Foundation

~\$7,500